

AI Idea Forge

Product Requirements Document (PRD)

Product Type: AI-powered validation and research platform

Status: Final Capstone Project

Live Application: discovery-agent-lab.lovable.app

1. Product Overview

AI Idea Forge is an AI-powered validation and research platform designed to transform a business idea or product concept into structured, actionable insights.

The platform uses six specialized AI agents to analyze market demand, competition, market gaps, feasibility, risks, and strategic opportunities. Results are presented through the Discovery Galaxy interface and downloadable reports.

2. Problem Statement

Evaluating a business idea or product concept traditionally requires research across market demand, trends, competitors, customer needs, market gaps, feasibility, risks, and strategic opportunities. This process can be time-consuming and difficult to organize.

AI Idea Forge provides a structured workflow that automates these analysis areas using specialized AI agents.

3. Product Goal

Provide users with a structured 360-degree view of an idea or product concept before making important business decisions.

The system should automate research and analysis, separate responsibilities across specialized agents, store individual results, provide an overall confidence score, visualize the analysis process, and generate a consolidated report.

4. Target Users

Primary: Startup founders, entrepreneurs, product innovators, business analysts, and individuals evaluating new product ideas.

Secondary: Students working on startup projects, innovation teams, product teams, and business consultants.

5. Core User Journey

User → Submit Idea / Product Concept → Frontend Validation → Supabase → n8n Workflow → Six AI Agents → Individual Results Stored → Confidence Score Calculation → Analysis Completed → Discovery Galaxy → Dashboard / Report.

6. Functional Requirements

FR-01 — User Authentication: Users can sign up, sign in, access analysis records, view profiles, and view roles. Roles are User and Admin.

FR-02 — Submit New Analysis: Users can submit a business idea or product concept, including title, description, and industry.

FR-03 — Store Analysis Request: The submitted idea is stored in Supabase and associated with the authenticated user. An `analysis_request_id` is used throughout the workflow.

7. AI Agent Requirements

Market Research: Evaluates market demand, industry trends, customer segments, market size and related market factors.

Competitor Analysis: Identifies direct and indirect competitors and evaluates the competitive landscape.

Gap Analysis: Identifies market gaps, unmet customer needs and innovation opportunities.

Feasibility Analysis: Evaluates business feasibility, technical complexity, resources, timeline and budget.

Risk & Challenge: Identifies market, technical, business, financial, legal and operational risks.

Recommendation Engine: Generates strategic recommendations and actionable next steps.

Each agent specializes in one business domain rather than asking a single AI model to perform the complete analysis.

8. AI Orchestration

n8n coordinates the workflow: receive the analysis request, fetch analysis data, send relevant information to the AI agent, receive structured JSON, parse the response, save the result to Supabase, continue the workflow, update analysis status, and complete the overall analysis.

9. Competitor Analysis Requirement

Competitor identification is based on the actual product/service and customer problem rather than simply the industry. Direct competitors are commercial companies, SaaS platforms, software products, startups, or services offering highly similar solutions. Indirect competitors are alternative commercial products, substitute technologies, adjacent solutions, or manual approaches customers may use instead.

10. Confidence Score

Each AI agent produces a `confidence_score`. The platform uses the six agent confidence scores to calculate the overall confidence value, stored in `ai_analysis_data.confidence_score`. The existing `matching_score` field remains unchanged.

Market Research + Competitor Analysis + Gap Analysis + Feasibility Analysis + Risk & Challenge + Recommendation Engine → Overall Confidence Score.

11. Re-analysis

When a user selects retry or re-analysis, the analysis workflow runs again, the six agent results are updated, `ai_analysis_data.reanalysis_count` increases by 1, and the confidence score is recalculated.

12. Discovery Galaxy

The Discovery Galaxy provides AI agent visualization, analysis progress, agent status, confidence score, analysis completion state, and an interactive user experience. Its purpose is to make the AI analysis process more transparent.

13. Dashboard

The dashboard provides analysis history, analysis status, overall confidence, confidence trends, re-analysis information, access to individual analysis results, and access to reports.

14. Admin Requirements

The system supports admin and user roles. Admin users can select View All to view all analysis records. Normal users can view their own analysis records.

15. Database Requirements

profiles: User profile and role information.

new_analysis_data: Submitted startup idea/product information.

ai_analysis_data: Master analysis record, status, confidence score and metadata.

market_research: Market research results.

competitor_analysis: Competitor analysis results.

gap_analysis: Gap analysis results.

feasibility_analysis: Feasibility results.

risk_challenge: Risk and challenge results.

recommendation_engine: Strategic recommendations.

16. Data Architecture

profiles → new_analysis_data → ai_analysis_data → six specialist analysis tables. The specialist tables are linked to the master analysis through the analysis request identifier.

17. Technology Requirements

Lovable: Frontend and user interface.

n8n: Webhooks, workflow orchestration, AI agent execution, data synchronization, error handling and status updates.

Groq: LLM processing.

Supabase: Authentication, user profiles, analysis records, agent results, confidence scores and re-analysis counts.

18. Non-Functional Requirements

Usability: Simple workflow from idea submission to final analysis.

Maintainability: Each AI agent remains independently maintainable.

Scalability: Architecture can support additional agents or analysis dimensions.

Reliability: Handle AI response errors, database errors, invalid JSON and workflow failures.

Security: Sensitive credentials and API keys must not be committed to GitHub.

19. Error Handling

AI Agent → Success → Parse JSON → Save Result; AI Agent → Error → Error Response. The workflow should provide appropriate handling for AI and database failures.

20. Reporting

Users can access structured results covering market research, competitors, gaps, feasibility, risks, recommendations and overall confidence through the application and downloadable reports.

21. Success Criteria

User can register/login; submit an idea/product concept; store the request in Supabase; trigger n8n; execute all six AI agents; store individual results; calculate overall confidence; update status; view Discovery Galaxy; view dashboard history; increment re-analysis count; allow admins to view all records; and access the final analysis/report.

22. Out of Scope

Additional AI agents; complex recommendation engines beyond the existing agent; advanced external research integrations; additional database tables unless required by existing functionality; major redesign of existing application pages; and additional reference documentation beyond required capstone deliverables.

23. Final Deliverables

Working AI Idea Forge application; n8n AI orchestration workflows; Supabase database architecture; AI agent prompts; system documentation; GitHub repository; project demonstration video; final LinkedIn post.

24. Product Value Proposition

AI Idea Forge helps users evaluate a business idea or product concept systematically by combining six specialized AI analyses into one automated research and validation workflow.

The key differentiator is not simply using AI to generate text, but orchestrating multiple specialized AI agents around a structured business-validation process.